



Astronomic Weekly Journal

Week 7: Mount Fundamentals and Tracking Stability

Field Notes | Mechanical Precision and Equatorial Drive

2026-04-29 | v1.0

DOI: 10.5281/zenodo.20110561

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Abstract

The equatorial mount is the mechanical foundation of any astrophotography system, responsible for counteracting Earth's rotation with sufficient precision that starlight accumulates without trailing across the imaging sensor during each sub-frame exposure. This journal presents a comprehensive engineering analysis of equatorial mount mechanics, performance classification, and selection methodology. Eight equations are derived from first principles, established astronomical references, software documentation, and clearly labelled engineering approximations: Equations (1) and (2) formalise the sidereal tracking rate and periodic-error pixel displacement; Equations (3) through (6) establish an approximate mount PSF contribution framework, expressing guided tracking residuals as FWHM components on independent right ascension (RA) and declination (DEC) axes and deriving a star elongation indicator; Equations (7) and (8) model the approximate unguided trail displacement versus sub-frame duration and identify a crossover regime where open-loop tracking may produce less RA displacement than the guided noise floor. Nine tables classify mounts by geometry, drive technology, performance tier, and representative market examples, and synthesise these into a selection guide keyed to image scale. Improved polar alignment is identified as the highest-leverage operational variable for Class I and II mounts, substantially reducing DEC tracking residual without hardware change, though mechanical backlash and other DEC error sources remain.

Keywords: equatorial mount; sidereal rate; periodic error; polar alignment; autoguiding; worm gear drive; harmonic drive; direct drive; tracking stability; image scale; pixel displacement; FWHM_{mount}; star roundness; DEC backlash; crossover time; mechanical resilience; mount selection; astrophotography; lucky imaging

Nomenclature

Symbol	Definition
ω_{sid}	Sidereal angular velocity at the celestial equator (arcsec s^{-1}); $\omega_{\text{sid}} = 2\pi / T_{\text{sid}} = 1296000 \text{ arcsec} / 86164.1 \text{ s} = 15.041 \text{ arcsec s}^{-1}$
ω_{eff}	Declination-adjusted effective sidereal drift rate; $\omega_{\text{eff}} = \omega_{\text{sid}} \times \cos(\delta)$ (arcsec s^{-1})
T_{sid}	Sidereal day; period of Earth's rotation relative to distant stars; $T_{\text{sid}} = 86164.1 \text{ s}$ (23 h 56 min 4.1 s mean solar time)
δ	Target declination (degrees); the factor $\cos(\delta)$ reduces the effective RA tracking demand toward the celestial poles
A_{PE}	Periodic error amplitude; one-half of the quoted peak-to-peak PE value (arcsec)
δ_{px}	Periodic error pixel displacement on the imaging sensor; $\delta_{\text{px}} = A_{\text{PE}} / s$ (px)
k	Pixel displacement tolerance fraction; the accepted fraction of one pixel for round-star imaging (dimensionless); typically $k = 0.5$, giving $A_{\text{PE,max}} = k \times s$
s	Image scale of the telescope-camera system (arcsec px^{-1}); $s = 206.265 \times p / f_{\text{mm}}$
p	Pixel pitch of the imaging sensor (μm)
f_{mm}	Effective focal length of the optical system (mm)
206.265	Plate-scale conversion constant; $206265 \text{ arcsec rad}^{-1} / 1000 (\mu\text{m to mm}) = 206.265 \text{ arcsec mm } \mu\text{m}^{-1}$
σ_{RA}	Guided RA tracking residual; 1- σ root-mean-square (RMS) of the guide star RA position deviation from the lock point, reported by PHD2 per session (arcsec). Empirical quantity; not derivable from specs.
σ_{DEC}	Guided DEC tracking residual; 1- σ RMS of the guide star DEC position deviation from the lock point, reported by PHD2 per session (arcsec). Typically larger than σ_{RA} due to DEC backlash.
ϵ_{total}	Total 2D guided tracking residual; magnitude of the RMS error vector; $\epsilon_{\text{total}} = \sqrt{\sigma_{\text{RA}}^2 + \sigma_{\text{DEC}}^2}$; reported as total RMS by PHD2 (arcsec)
FWHM _{mount}	Mount PSF contribution; full-width at half-maximum (FWHM) of the Gaussian tracking-error PSF added by the mount to the delivered star profile. RA and DEC axes are independent: $\text{FWHM}_{\text{RA, mount}} = 2.35 \times \sigma_{\text{RA}}$; $\text{FWHM}_{\text{DEC, mount}} = 2.35 \times \sigma_{\text{DEC}}$; equivalent circular: $\text{FWHM}_{\text{mount, eq}} = 2.35 \times \sqrt{(\sigma_{\text{RA}}^2 + \sigma_{\text{DEC}}^2)/2} = (2.35/\sqrt{2}) \times \epsilon_{\text{total}}$ (arcsec)



Symbol	Definition
r	DEC/RA tracking residual ratio; $r = \sigma_{\text{DEC}} / \sigma_{\text{RA}}$ (dimensionless). $r = 1.0$ for isotropic guiding (perfect polar alignment); $r > 1.0$ when DEC backlash or polar drift is present. Per-class worst-case values: Class I $r = 2.0$; II $r = 1.5$; III harmonic $r = 1.2$; IV $r = 1.15$; V $r = 1.0$
RND	Star roundness metric; $\text{RND} = (r - 1) / (r + 1)$ (dimensionless). $\text{RND} = 0$: perfectly round stars. $\text{RND} > 0.15$ (15 percent): elongation is generally perceptible in the final image
δ_{trail}	Approximate RA displacement produced by sinusoidal PE during an unguided sub-frame of duration t starting at the PE zero-crossing; $\delta_{\text{trail}}(t) = A_{\text{PE}} \times \sin(2 \pi \times t / T_w)$ for $t \leq T_w/4$ (arcsec). Phase-specific approximation; not a general FWHM model.
t_c	Crossover time; sub-frame duration below which the simplified unguided RA displacement estimate (δ_{trail}) remains below the assumed guided RA displacement floor ($2.35 \times \sigma_{\text{RA}}$); $t_c = (T_w / 2 \pi) \times \arcsin(2.35 \times \sigma_{\text{RA}} / A_{\text{PE}})$ (s)
T_w	Worm rotation period; time for one complete revolution of the RA worm shaft; sets the PE repetition period (s). Typical values: 480 s for HEQ5-class mounts

1. Introduction

Among all the subsystems in an astrophotography platform, the equatorial mount sets the boundary conditions for every other component. The optical system defines what angular resolution is physically attainable; the mount determines whether that resolution is realised or lost to tracking error during each exposure. A mount that fails to hold a star steady to within a fraction of a pixel commits the limiting error at the moment of photon collection, before any processing can help.

Earth's rotation rate relative to distant stars is fixed by celestial mechanics at 15.041 arcseconds per second at the equator. Any mechanical deviation from this rate drifts the star image across the sensor in proportion to elapsed time. Converting that angular drift into pixel displacement requires only the image scale of the optical system. The governing equations for this conversion, derived in this journal, are instrument-independent and apply to any mount-telescope-camera combination for which the periodic error specification and image scale are known.

The journal proceeds from mount geometry and drive technology through the mathematical tolerance framework to a ten-

configuration trade study. Polar alignment methods and autoguiding are examined within the tolerance framework established by the derived equations, and a forward-looking section connects mount performance to the camera sampling discussion that follows in Week 8.

2. The Equatorial Mount: Geometry and Function

Astronomical mounts are built around one of two fundamental axis geometries. An altitude-azimuth (alt-az) design orients its two rotation axes to the local vertical and horizontal, making it mechanically simple and intuitive for visual observation.

Tracking a star with an alt-az mount requires both axes to move simultaneously at continuously varying rates, and the geometry introduces field rotation: the sky rotates relative to the sensor during each exposure, degrading sub-frames in many amateur deep-sky cases, often limiting them to tens of seconds, with the exact limit depending on target altitude, azimuth, focal length, sensor size, and accepted star elongation.

Alt-az designs therefore serve visual observation, electronically assisted astronomy (EAA), and planetary and lunar video imaging.



Table 1. Principal mount geometry classes in current amateur and professional use. Field rotation limits long-exposure sub-frame duration for alt-az designs unless a field derotator is installed. All equatorial geometry classes eliminate field rotation by aligning the primary tracking axis with Earth's rotation axis. This is a descriptive taxonomy derived from the literature and current market; no formal standards body has defined these categories.

Geometry class	Axis design	Field rotation	Primary observing domain
Manual alt-az	Altitude + azimuth axes; no motor	N/A (no tracking)	Naked-eye visual; beginners; outreach
GoTo alt-az	Both axes motorised; computer-driven at variable rates	Yes; limits subs to approximately 15-30 s without derotator	Visual; EAA; planetary; lunar
Dobsonian	Alt-az rocker base; manual push-to or motorised	Yes during tracking	Large-aperture visual astronomy
Camera tracker (single-axis EQ)	RA axis only; no Dec motor; camera or lens payload only	No	Wide-field DSLR and mirrorless imaging; Milky Way; lens imaging
German equatorial (GEM)	Polar axis (RA motor) + Dec axis (guide port); counterweight-balanced	No	Deep-sky astrophotography; all telescope imaging
Centre-balanced equatorial (CEM)	Polar axis (RA motor) + Dec axis; Z-shaped centre-balanced design	No	Deep-sky astrophotography; higher payload-to-mount-weight ratio than GEM
Fork-on-wedge (EQ)	Alt-az fork mount tilted to local latitude via equatorial wedge	No (when on wedge)	SCT long-exposure imaging; avoids meridian flip
Direct drive (alt-az or EQ)	Both axes driven by magnetic coils; absolute encoders; no gears	No (EQ); encoder-corrected (alt-az)	Professional observatory; research; satellite and space debris tracking

An equatorial mount eliminates field rotation by inclining its primary axis parallel to Earth's rotation axis; a single motor driving that polar axis at the sidereal rate keeps any celestial target stationary at the focal plane for the full sub-frame duration. Table 1 classifies the principal mount geometry classes available in the current market, their tracking approach, and the observing domains each serves.

Within the equatorial geometry classes, the drive technology determines the magnitude and character of the residual periodic error. Different drive mechanisms produce distinctly different PE signatures that affect the difficulty of autoguiding correction and the unguided sub-frame ceiling. Table 2 classifies the drive mechanisms available across the current market.

Within the equatorial geometry family the German equatorial mount is the dominant design in amateur astrophotography. A GEM consists of two perpendicular rotation axes: the polar axis, inclined to the horizontal plane at the observer's geographic latitude and aligned parallel to Earth's rotation axis; and the

declination axis, mounted perpendicular to the polar axis at its top end.

The optical tube assembly attaches to one end of the declination axis, with a counterweight rod extending from the opposite end to balance the load. When the polar axis is driven at the sidereal rate, any target on the celestial sphere remains stationary at the focal plane for the full duration of the exposure. The ZWO AM7 adopts the same polar-and-declination geometry but replaces the counterweight with a harmonic drive that provides sufficient holding torque at any position angle, eliminating counterweight mass while retaining identical tracking geometry.

Within the equatorial mount market, a de facto stratification has emerged based on payload capacity and tracking precision. No formal standards body has defined these tiers, but the stratification is consistent enough across manufacturers to serve as a practical selection framework. Table 3 presents this tier structure, with each class defined by its payload range, drive type, and typical tracking precision.



Table 2. Drive technology classification for equatorial and alt-az mounts. Values labelled (M) are from manufacturer-published specifications. Values labelled (NP) are not formally published; community measurement data confirm the qualitative ranking but specific figures vary by unit. Harmonic drive PE depends strongly on payload and pointing direction; manufacturer RMS figures represent best-case conditions and real-world peak displacements will exceed the quoted RMS by a factor of $\sqrt{2}$ or more for a sinusoidal signal component.

Drive technology	PE character	Typical unguided PE
Manual friction (e.g., Dobsonian)	None	N/A
Single-axis worm (camera trackers)	Sinusoidal; single period; RA axis only	Not formally published (NP); small worm wheel limits precision
Standard worm gear (both axes)	Sinusoidal; ~8-10 min period; predictable; correctable via PEC	20-80 arcsec peak-to-peak
Belt-assisted worm gear	Sinusoidal; belt reduces gear harmonics and backlash	Typically lower than standard worm; not formally published (NP)
Harmonic drive (strain-wave gear)	Multi-frequency; payload-dependent; less predictable than worm	<5 arcsec RMS manufacturer best-case (M); real-world varies with load
Precision worm (lapped, matched pairs)	Sinusoidal; very smooth; highly repeatable for PEC training	Typically +-2 to +-8 arcsec peak-to-peak (M); <2 arcsec after PEC (M)
Direct drive + absolute encoders	Encoder-eliminated; no mechanical gear cycle	<0.25 arcsec RMS unguided (M); effectively zero with active control

Across all equatorial classes, three mechanical error sources govern imaging performance. Tracking is the open-loop rotation of the RA axis at the commanded sidereal rate. Guiding is closed-loop correction via a guide camera that issues corrective pulses during the exposure. Polar alignment error is a systematic offset between the RA axis and Earth's rotation axis, producing declination drift that neither tracking nor guiding can eliminate without mechanical adjustment. The pixel-plane consequence of any of these three error sources depends on image scale, the

relationship formalised in Equation (2).

- Tracking error and polar alignment error are reduced before the session begins by careful mechanical setup; they cannot be recovered in post-processing.
- Guiding residual is managed in real time during the session; its magnitude depends on guide optics, guide camera sensitivity, atmospheric seeing, and the mount's response to correction pulses.
- Periodic error is a property of the drive mechanism and repeats predictably over the

Table 3. De facto performance tier classification for equatorial mounts. Payload values represent instrument capacity excluding counterweights. The recommended imaging load is no more than two-thirds of the rated capacity. Values labelled (M) are manufacturer-published; (NP) are not formally published. Class III encompasses two distinct drive sub-types with different PE characteristics; the harmonic drive RMS value is a manufacturer best-case figure. Class boundary payloads overlap because individual models within each tier vary; the drive type and PE specification are the defining properties, not payload alone.

Class	Typical payload (kg)	Drive type	Typical PE (unguided)	Guiding status
0 Camera tracker	2-5	Single-axis worm; RA only	Not published (NP)	Strongly recommended above 135 mm focal length
I Entry GEM / CEM	8-14	Standard worm gear	20-80 arcsec peak-to-peak	Essential for all telescope imaging
II Mid-range GEM / CEM	13-22	Standard or belt-assisted worm	10-30 arcsec peak-to-peak	Essential
III Upper-mid GEM / CEM	20-35	Belt-assisted worm or harmonic drive	Worm: typically lower (NP); Harmonic: <5 arcsec RMS (M)	Essential
IV Premium GEM	25-55	Precision lapped worm (+/- encoder option)	+2 to +-8 arcsec p-p (M); <2 arcsec after PEC (M)	Strongly recommended
V Observatory / direct drive	45-500+	Precision worm + absolute encoder, or direct drive	<1 arcsec with pointing model; ~0 for direct drive (M)	Optional for wide-field unguided work



worm cycle; it is the primary driver of the mount specifications in Table 5.

- Angular mount error is specified in arcseconds; imaging consequences are measured in pixels. The conversion between these units is image scale, quantified by Equation (2) and applied across all ten configurations in Table 9.

2.1 Mount Selection: A Decision Framework

Tables 1 through 4 provide the basis for a systematic mount selection process. The five-step framework below applies the classification to any observing goal, from a first tracking mount for Milky Way imaging to a permanent observatory installation.

1. Determine the observing domain. Select a geometry class from Table 1. Visual observers can choose alt-az GoTo designs; wide-field DSLR photographers should consider a camera tracker (Class 0); deep-sky imaging through a telescope requires an equatorial GEM, CEM, or fork-on-wedge (Classes I-V).
2. Calculate the required payload. Sum the masses of the optical tube, imaging camera, filter wheel, guide scope or off-axis guider, guide camera, and all accessories. Apply the two-thirds rule: the total imaging load should not exceed two-thirds of the mount's rated payload capacity for sessions where tracking precision is the primary constraint.
3. Derive the PE tolerance from the planned image scale. Using Equation (2), compute the 0.5-pixel budget for the telescope-camera combination: $A_{PE_max} = 0.5 \times s$, where $s = 206.265 \times p / f_{mm}$. Compare this tolerance to the PE values in Table 4. The minimum class where the published PE does not exceed A_{PE_max} will deliver

round stars without guiding if all other error sources are negligible; in practice, autoguiding is required for all configurations in Table 9 at the PE specifications available in this instrument collection.

4. Account for portability. If the mount must be transported to observing sites, total system weight is a practical constraint independent of payload. Class 0 and Class I systems typically total under 12 kg; Class II systems 15-25 kg; Class III and above are generally semi-permanent or permanent installations.
5. Verify site and infrastructure compatibility. Classes IV-V benefit from a concrete pier, precise azimuth control, and in the case of direct-drive Class V systems, a dedicated computer for encoder-based motor control. Field observers using Classes 0-III require a portable power source rated for the mount's current draw and a stable, level tripod surface.

2.2 Polar Axis Alignment

Polar alignment orients the mount's RA axis to coincide with Earth's rotation axis. For an observer at latitude -37.9 degrees in southern Victoria, the south celestial pole lies approximately 37.9 degrees above the southern horizon, near the faint star Σ Octantis. A misaligned polar axis means the RA drive cannot fully compensate for Earth's rotation, and the star traces an arc across the sensor rather than remaining stationary. The resulting declination drift depends on both the misalignment angle and the hour angle of the target, and is independent of the drive's periodic error [2].



Table 4. Illustrative market examples by performance tier. Payload values are rated instrument capacity excluding counterweights as published by manufacturers; a conservative imaging load limit is approximately two-thirds of the rated value for long-exposure work, though this is an empirical rule rather than a manufacturer guarantee. PE values labelled (M) are manufacturer-published; (I) are independent community measurements not formally published; (NP) indicates no figure is published by any source. RMS, peak, and peak-to-peak specifications use different conventions and must not be directly compared. The three Austronomic instrument collection mounts are EQM-35 (Class I), HEQ5-R Pro (Class II), and ZWO AM7 (Class III harmonic). All specifications are compiled from manufacturer documentation and community sources as described in the code comments; manufacturer datasheets are the authoritative source for any procurement decision. ZWO AM7 payload and PE specifications are from current official ZWO product documentation [7].

Class	Representative mount	Rated payload (kg)	Published PE
0	Sky-Watcher Star Adventurer 2i	5	Not published (NP)
0	Sky-Watcher Star Adventurer GTi	5	Not published (NP)
0	iOptron SkyGuider Pro	5	Not published (NP)
0	iOptron SkyTracker Pro	1.5	Not published (NP)
0	Vixen Polaris U	2	Not published (NP)
I	Sky-Watcher EQM-35	10	~15-25 arcsec peak-to-peak (I)
I	iOptron CEM26	12	<+/-10 arcsec (M)
I	iOptron CEM26EC	12	<+/-3 arcsec with EC encoders (M)
I	Celestron Advanced VX	14.5	Not published (NP); belt-driven worm
I	Bresser Exos-2 GoTo	10	Not published (NP)
II	Sky-Watcher HEQ5-R Pro	13.7	Typically +/-10 arcsec (I); not formally published
II	iOptron CEM40	18	+/-7 arcsec (M)
II	iOptron GEM45	20	Not published (NP)
II	Celestron CGX	22.7	Not published (NP)
II	Orion Atlas EQ-G	18	Not published (NP); Synta OEM, community data comparable to HEQ5-R
III	Sky-Watcher EQ6-R Pro	20	Not published (NP); belt drive reduces harmonic content
III	iOptron CEM70	31.8	+/-3.5 arcsec (M); <0.3 arcsec RMS with EC encoder option (M)
III	ZWO AM5	13 (no cw) / 20 (with cw)	Controlled within +/-10 arcsec (M); individual PE test report supplied
III	ZWO AM7	20 (no cw) / 30 (with cw)	Controlled within +/-10 arcsec (M); individual PE test report supplied
III	Pegasus NYX-101	15	Not published (NP); harmonic drive
IV	Losmandy G11	27	+/-5 arcsec or better (M)
IV	10Micron GM1000 HPS	25	+/-4 arcsec max mechanical (M); +/-1.5 arcsec with motor control (M)
IV	Software Bisque Paramount MX+	45	7 arcsec peak-to-peak before PEC (M); <1 arcsec after PEC (M)
IV	Takahashi EM-200 Temma 3	20	Not published (NP); precision lapped worm; community data <+/-5 arcsec
IV	Astro-Physics Mach2 GTO	36	Not published (NP); high-precision worm; community data <+/-3 arcsec
IV	iOptron CEM120	50	+/-5 arcsec (M)
V	ASA DDM100 (direct drive)	100	<0.25 arcsec RMS unguided (M)
V	PlaneWave L-350 (direct drive)	45	Zero PE; encoder-eliminated (M)
V	PlaneWave L-500 (direct drive)	100	Zero PE; encoder-eliminated (M)
V	10Micron GM2000 HPS	80	+/-4 arcsec max mechanical (M); encoder feedback reduces to <1 arcsec
V	Software Bisque Paramount ME II	68	7 arcsec peak-to-peak before PEC (M); <1 arcsec after PEC (M)

For guided imaging sessions, polar alignment errors of a few arcminutes are routinely suppressed by the guiding loop without visible effect on star shape, because the drift rate from a small polar error is slow relative to the guide update frequency. For unguided imaging, the accumulated declination drift over the full sub-frame duration must remain within the pixel

tolerance criterion from Equation (2). Electronic polar alignment routines, which compare stellar drift between two positions to compute the residual polar error vector, can achieve alignments of 2 to 10 arcminutes in the field and are the practical method for session preparation with all three mounts in this series [2].



2.3 Mechanical Balance, Payload, and Meridian Flip

The rated payload defines the maximum imaging load on the optical side of the polar axis. Operating near the payload limit increases mechanical stress on the drive components, degrades pointing repeatability, and worsens the periodic error performance relative to the factory specification. The two-thirds rule from the selection framework applies here directly. Precise balance in both RA and declination, established before the session begins, reduces the torque demand on the drive and maintains the worm mesh within its design load range for worm-gear mounts, or holds the harmonic flexspline within its rated deflection for the ZWO AM7.

A constraint specific to the GEM geometry is the meridian flip. When the optical tube crosses the meridian during a long tracking run, the telescope must be rotated approximately 180 degrees in hour angle to continue following the same target on the opposite side. This interrupts the imaging session, requires recentring the target and recalibrating the guide system, and introduces a data gap of several minutes. Selecting a target that transits well after the planned session start allows the full imaging window to fall on one side of the meridian, avoiding the flip entirely. With the geometric and balance framework established, the analysis turns to the dominant residual tracking error from the drive itself.

3. Drive Technology and Periodic Error

The tracking accuracy of an equatorial mount is bounded not only by the quality of the polar alignment but by the mechanical fidelity of its

drive system. Both worm-gear and harmonic-drive mounts introduce a periodic, repeating pattern of angular rate variation that manifests as a sinusoidal oscillation of the star position in the RA direction. This oscillation, known as periodic error (PE), is the dominant residual tracking imperfection in a well-aligned mount and the primary engineering parameter distinguishing mount grades for astrophotography.

3.1 Worm Gear Drives

A worm gear drive consists of a helical worm shaft meshing with a large-diameter worm wheel attached to the RA axis. One full worm revolution advances the wheel by exactly one tooth, producing a large mechanical advantage and high angular resolution. Imperfections in the worm tooth profile, the eccentricity of the worm wheel on its axis, and variation in meshing pressure all introduce a periodic angular position error that repeats once per worm revolution. For the Sky-Watcher HEQ5-R Pro, the worm period is approximately 8 minutes per manufacturer documentation; the EQM-35 uses a similar worm-gear architecture and a period in the same range is typical, but the exact value for any specific unit is most reliably determined from its RA drift log with corrections disabled [2].

Periodic error in worm-gear mounts is quoted as peak-to-peak amplitude in arcseconds over one worm cycle. A specification of 30 arcseconds peak-to-peak means the star's RA position oscillates between +15 and -15 arcseconds from the mean tracking line over one worm period. This amplitude is independent of the attached optical system: the same mount produces the same PE in arcseconds regardless of focal length. The



consequence in pixels depends entirely on the image scale of the telescope-camera pairing, formalised in Section 4.

3.2 Harmonic Drive Technology

A harmonic drive, or strain-wave gear, replaces the worm and worm wheel with three concentric components: a flexible ring called the flexspline, deformed into an ellipse by a rotating inner bearing called the wave generator, against the teeth of a rigid outer circular spline. The tooth engagement zone circulates continuously around the ring, producing motion that is mechanically smooth at every rotational position. The absence of a discrete, repeating meshing cycle removes the dominant mechanism that generates worm-period PE in gear-driven mounts, which accounts for the substantially lower PE specifications typical of harmonic drive designs.

Current ZWO official documentation for the AM7 states that periodic error is controlled within ± 10 arcseconds peak, with an individual PE test report supplied for each unit. This is a peak-bounded specification, not an RMS value. For a pure sinusoidal component, RMS multiplied by $\sqrt{2}$ gives the sinusoidal peak amplitude; however, harmonic-drive tracking error is not necessarily a pure sinusoid, it is multi-frequency and payload-dependent. The actual peak excursion therefore cannot be reliably estimated from an RMS figure by a fixed conversion factor; it must be read from the unit's individual PE curve or a live guide log. RMS, peak, and peak-to-peak values must not be directly compared without stating the conversion convention.

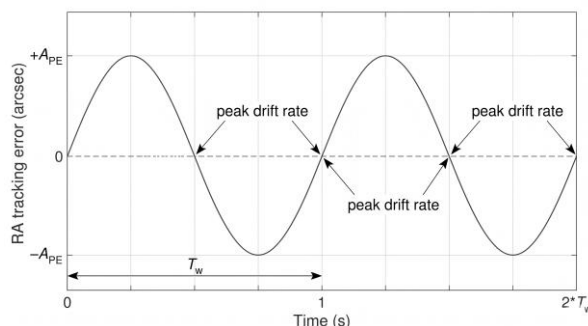


Figure 1. Representative periodic error signature of a worm-gear equatorial mount. The amplitude A_{PE} is the input to Equation (2) for computing pixel-plane star displacement. The zero-crossings mark the instants of maximum drift rate within the cycle.

Unlike worm-gear PE, harmonic-drive error is less amenable to periodic error correction (PEC) training because its error content is not repeatable at a fixed worm period.

Figure 1 depicts a representative periodic error signature for a worm-gear equatorial mount, showing the sinusoidal RA tracking error across two complete worm cycles with key measurement parameters labelled.

4. Tracking Mathematics

Two governing equations organise the quantitative analysis of mount tracking performance. The first derives the absolute angular velocity that every equatorial drive must reproduce to keep the celestial sphere stationary at the focal plane; the second converts mount periodic error from its native arcsecond specification into pixel units that directly determine whether a given configuration can produce round, untrailed star images.



4.1 Sidereal Angular Velocity

The sidereal day is the period of Earth's rotation with respect to distant stars, distinct from the solar day, which adds the daily increment of Earth's orbital motion around the Sun. Equation (1) gives the sidereal angular velocity at the celestial equator: two pi radians divided by the sidereal period, converted to the arcsecond-per-second units used in mount PE specifications. The sidereal day is $T_{\text{sid}} = 86164.1$ seconds, determined from very-long-baseline interferometric measurements of Earth's rotation rate [1].

$$\omega_{\text{sid}} = \frac{2\pi}{T_{\text{sid}}} \quad (1)$$

ω_{sid} = Sidereal angular velocity at the celestial equator (arcsec s^{-1}); evaluates to 15.041 arcsec s^{-1}

T_{sid} = Sidereal day; $T_{\text{sid}} = 86164.1$ s (23 h 56 min 4.1 s mean solar time)

ω_{eff} = Declination-adjusted effective sidereal rate; $\omega_{\text{eff}} = \omega_{\text{sid}} \times \cos(\delta)$ (arcsec s^{-1})

Substituting $T_{\text{sid}} = 86164.1$ s gives $\omega_{\text{sid}} = 15.041$ arcseconds per second at the celestial equator. At a target declination δ , the effective RA drift rate falls by the factor $\cos(\delta)$, because hour-angle circles converge toward the poles. At $\delta = -60$ degrees, a common southern deep-sky declination, $\cos(\delta) = 0.50$, halving the tracking demand relative to the equatorial rate. The declination-adjusted effective velocity is $\omega_{\text{eff}} = \omega_{\text{sid}} \times \cos(\delta)$, derived from Equation (1) [4].

4.2 Periodic Error Pixel Displacement

The tolerance criterion for mount periodic error is most usefully expressed in pixels rather than arcseconds, because the imaging system determines what magnitude of displacement is visible as a star trail. Equation (2) converts the PE amplitude from its native arcsecond measure into a pixel count by dividing by the image scale of the telescope-camera system. The result is the maximum extent to which a star centroid will deviate from its mean position on the sensor during one worm cycle: the key quantity for predicting whether star images will be round or elongated.

$$\delta_{\text{px}} = \frac{A_{\text{PE}}}{s} \quad (2)$$

δ_{px} = Periodic error pixel displacement; maximum star centroid deviation on the sensor during one worm cycle (px)

A_{PE} = Periodic error amplitude; half of the quoted peak-to-peak PE value (arcsec)

s = Image scale of the telescope-camera system (arcsec px^{-1}); $s = 206.265 \times p / f_{\text{mm}}$

k = Pixel displacement tolerance fraction (dimensionless); typically $k = 0.5$; rearranging gives $A_{\text{PE}_{\text{max}}} = k \times s$ as the maximum allowable amplitude

Applying the relationship: an A_{PE} of 15 arcseconds displaces a star by 12.3 pixels at 1.22 arcsec/px, and by 39.5 pixels at 0.38 arcsec/px, a factor of more than three from the optics alone with the mount's mechanical performance unchanged. Rearranging gives $A_{\text{PE}_{\text{max}}} = k \times s$, so the 0.5-pixel budget column in Table 9 (computed as $0.5 \times s$) is a direct readout of the maximum tolerable



Table 5. Equatorial mount specifications. PE values are manufacturer-specified. Worm-gear mounts (EQM-35, HEQ5-R Pro) report peak-to-peak amplitude over one worm cycle. The ZWO AM7 specification is stated as periodic error controlled within ± 10 arcsec peak, with an individual PE test report supplied for each mount. This is not a guided RMS specification and must not be compared directly with PHD2 RMS values. RMS, peak, and peak-to-peak values use different conventions and cannot be directly compared without stating the conversion assumption.

Mount	Drive type	Rated payload (kg)	Typical PE (arcsec)	PE metric
Sky-Watcher EQM-35	Worm gear GEM	10	30 to 60	Peak-to-peak
Sky-Watcher HEQ5-R Pro	Worm gear GEM	13.6	10 to 25	Peak-to-peak
ZWO AM7	Harmonic drive GEM	20 (no cw) / 30 (with cw)	Controlled within ± 10	Peak (individual PE test report supplied)

angular error from any source, derived from Equation (2).

Table 5 summarises the mechanical specifications of the three equatorial mounts in the instrument collection, providing the PE values used as input to Equation (2) in the trade study that follows.

Table 5 shows that the three mounts span a PE range differing by more than an order of magnitude: from the EQM-35's 30 to 60 arcseconds peak-to-peak to the HEQ5-R Pro's typical 10 to 25 arcseconds peak-to-peak, and the AM7's official ± 10 arcseconds peak specification with individual PE test report. These open-loop PE values are the input to Equation (2); the guided tracking performance of each mount is a separate, session-measured quantity not derivable from the PE specification alone.

5. Mount Performance Framework

This section develops the quantitative engineering framework for mount performance. Section 5.1 identifies the three independent classification families that govern mount performance for astrophotography. Sections 5.2 and 5.3 derive the approximate FWHM_{mount} equations for guided and unguided tracking respectively, including the crossover regime where open-loop tracking

may be preferable to closed-loop guiding. Section 5.4 analyses polar alignment as a session-level performance variable. Sections 5.5 and 5.6 complete the framework with a mechanical resilience classification and a mount selection matrix keyed to image scale.

5.1 Three Performance Families

Mount performance for astrophotography is governed by three independent classification families. Each family represents a distinct engineering dimension that cannot be substituted by the others: improving one family does not compensate for deficiency in another.

- Family 1 - Payload capacity [kg]: a feasibility gate that determines whether a given optical system can be physically mounted. The recommended imaging payload is no more than two-thirds of the rated visual capacity for sessions where tracking precision is the primary constraint.
- Family 2 - RA and DEC tracking accuracy [arcsec]: the primary imaging-quality family. RA accuracy is governed by periodic error (unguided) and guided RMS residual. DEC accuracy is governed by DEC backlash and polar alignment residual. Both axes produce independent FWHM contributions to the delivered star PSF, derived in Section 3.4.



- Family 3 - Mechanical resilience: a dynamic structural property governing the mount's response to wind loading, vibration from shutter actuation, and thermal deformation. Unlike Families 1 and 2, resilience cannot be reduced to a reliable closed-form equation; it is classified qualitatively in Section 3.7.

Payload and mechanical resilience are operational constraints that bound the feasible design space. Tracking accuracy (Family 2) is the engineering performance family that directly determines delivered image quality and is the subject of the quantitative framework developed in Sections 3.4 and 3.5.

5.2 The FWHM_{mount} Framework: Guided Tracking (Engineering Approximation)

When autoguiding is active, the guide loop measures the centroid position of a guide star at each frame interval and issues correction pulses to the mount. The residual tracking error, reported by PHD2 as separate RA and DEC root-mean-square values in arcseconds, characterises the precision of the closed-loop system. These residuals are measured empirically per session: they depend on mount mechanics, polar alignment quality, atmospheric seeing at the guide star, guide exposure duration, algorithm aggressiveness, and guide optics. They are not derivable from manufacturer specifications and vary between sessions at the same site. PHD2 RMS alone does not prove that the residual distribution is Gaussian, stationary, or directly transferable from guide-camera motion to imaging-camera PSF; the FWHM_{mount} framework below is therefore an engineering approximation for planning and diagnosis, not a guaranteed PSF model [5].

The tracking residuals in RA and DEC are statistically independent and If the guided residuals are approximated as Gaussian and independent, a working assumption reasonable for a well-tuned guide loop but not guaranteed, each axis contributes an approximate Gaussian PSF component to the delivered star profile. Equations (3) and (4) express this approximation for the RA and DEC axes respectively.

$$FWHM_{RA, mount} = 2.35 \times \sigma_{RA} \quad (3)$$

FWHM_{RA, mount} = Approximate RA axis mount PSF contribution; FWHM of the estimated Gaussian tracking-error distribution in RA (arcsec). Engineering approximation assuming Gaussian residuals.

σ_{RA} = Guided RA tracking residual; 1-σ RMS from PHD2 per session (arcsec)

$$FWHM_{DEC, mount} = 2.35 \times \sigma_{DEC} \quad (4)$$

FWHM_{DEC, mount} = DEC axis mount PSF contribution; FWHM of the Gaussian tracking error distribution in DEC (arcsec)

σ_{DEC} = Guided DEC tracking residual; 1-σ RMS from PHD2 per session (arcsec). Typically σ_{DEC} ≥ σ_{RA} due to DEC backlash and polar drift

Because the RA and DEC axes are perpendicular and independent, the resulting star PSF contribution from tracking is an elliptical Gaussian, not circular. A single equivalent circular FWHM is obtained by averaging the per-axis variances, which is the correct formulation for combining two independent Gaussian components. Equation (5) gives this equivalent circular FWHM_{mount}. It is important to note that this differs from simply multiplying PHD2's total



Table 6. Per-class guided tracking estimates and elongation indicator. This table is a planning model, not a specification table: it should be used to identify likely mount tiers for evaluation, not to guarantee suitability for a specific configuration. σ_{RA} values are median community-data estimates from PHD2 log aggregates; actual performance varies significantly by session, site, and setup. r_{worst} is the worst-case DEC/RA residual ratio per class, derived from backlash severity and typical polar alignment residuals. $FWHM_{mount,eq}$ is an engineering approximation from Equation (5) assuming Gaussian independent residuals; suitable for planning and diagnosis only, not a guaranteed PSF specification. RND from Equation (6) is a diagnostic heuristic; values above 0.15 warrant checking measured eccentricity in the delivered image. Good polar alignment reduces r toward 1.0 for any class, substantially reducing but not eliminating the DEC tracking residual (see Section 5.4).

Class	σ_{RA} (arcsec)	r_{worst}	σ_{DEC} worst (arcsec)	$FWHM_{mount,eq}$ worst (arcsec)	RND worst (%)	Drive / backlash basis
0 Camera tracker	N/A (no guiding)	N/A	N/A	N/A	N/A	Single-axis; no DEC motor
I Entry GEM	~1.15	2.0	~2.30	2.01	33% ELONGATED	Heavy backlash 5-30 arcsec; standard polar alignment
II Mid-range GEM	~0.75	1.5	~1.13	1.24	20% ELONGATED	Moderate backlash 2-15 arcsec
III Upper-mid (harmonic)	~0.55	1.2	~0.66	0.88	9% acceptable	Near-zero backlash; harmonic drive
IV Premium GEM	~0.30	1.15	~0.35	0.48	7% acceptable	Spring-loaded precision worm; minimal backlash
V Observatory / DD	~0.15	1.0	~0.15	0.25	0% round	Direct drive or encoder; zero backlash; DEC corrections minimal

RMS ($\epsilon_{total} = \sqrt{\sigma_{RA}^2 + \sigma_{DEC}^2}$) by 2.35, that formulation would overestimate the equivalent circular FWHM by a factor of $\sqrt{2}$.

$$FWHM_{mount,eq} = \frac{2.35}{\sqrt{2}} \sqrt{\sigma_{RA}^2 + \sigma_{DEC}^2} \quad (5)$$

$FWHM_{mount,eq}$ = Equivalent circular mount PSF contribution; circular approximation to the elliptical Gaussian tracking PSF (arcsec). Also written as $2.35 \times \sqrt{(\sigma_{RA}^2 + \sigma_{DEC}^2)/2}$

ϵ_{total} = Total 2D guided residual; $\epsilon_{total} = \sqrt{\sigma_{RA}^2 + \sigma_{DEC}^2}$ (arcsec); reported as total RMS by PHD2

When σ_{RA} differs from σ_{DEC} , the star image is elongated. The star roundness metric RND quantifies this elongation as a dimensionless ratio derived directly from the DEC-to-RA residual ratio $r = \sigma_{DEC} / \sigma_{RA}$. Equation (6) expresses RND in a form that is independent of the absolute residual magnitudes, only the ratio between the two axes determines the degree of elongation.

$$RND = \frac{r - 1}{r + 1} \quad (6)$$

RND = Internal elongation indicator (dimensionless); diagnostic heuristic only, not a formal standard. $RND = 0$: equal RA and DEC residuals (no directional preference). Values above 0.15 suggest measurable elongation and warrant checking measured eccentricity in the delivered image before use as an acceptance criterion.

r = DEC/RA tracking residual ratio; $r = \sigma_{DEC} / \sigma_{RA}$ (dimensionless). $r = 1.0$ when RA and DEC residuals are equal in RMS magnitude. Good polar alignment can reduce DEC drift and move r closer to 1.0, but r also depends on backlash, seeing, wind, stiction, and guide-loop behaviour. $r > 1.0$ typically indicates DEC residual exceeds RA residual.

Applying Equation (6): for $r = 1.5$ (Class II worm-gear mount with moderate backlash and typical polar alignment), $RND = (1.5 - 1)/(1.5 + 1) = 0.20$, corresponding to 20 percent elongation, above the 15 percent perceptibility



threshold. For $r = 2.0$ (Class I mount with heavy backlash), $RND = 0.33$: stars are visibly elongated even with guiding active. The 15 percent threshold corresponds to $r = 1.35$, regardless of the absolute σ values. Table 7 presents the per-class worst-case r values, RND , and $FWHM_{mount,eq}$ derived from Equation (6)

5.3 The Unguided Trail Displacement and the Crossover Regime

For open-loop sidereal tracking without a guide loop, sinusoidal PE causes the star to drift across the sensor during the exposure. The displacement depends on both the exposure duration and the PE phase at the start of the exposure. Equation (7) gives the approximate displacement for a short exposure beginning near the steepest part of the PE curve (zero-crossing, worst-case phase) and durations not exceeding one quarter of the worm period. This is a simplified, phase-specific approximation; it is not a general FWHM model and should not be used as a universal unguided exposure-limit equation. The true worst-case displacement over a longer interval may approach the full peak-to-peak PE range, not merely the amplitude.

$$\delta_{trail}(t) = A_{PE} \cdot \sin\left(\frac{2\pi}{T_w} t\right), \quad t \leq \frac{T_w}{4} \quad (7)$$

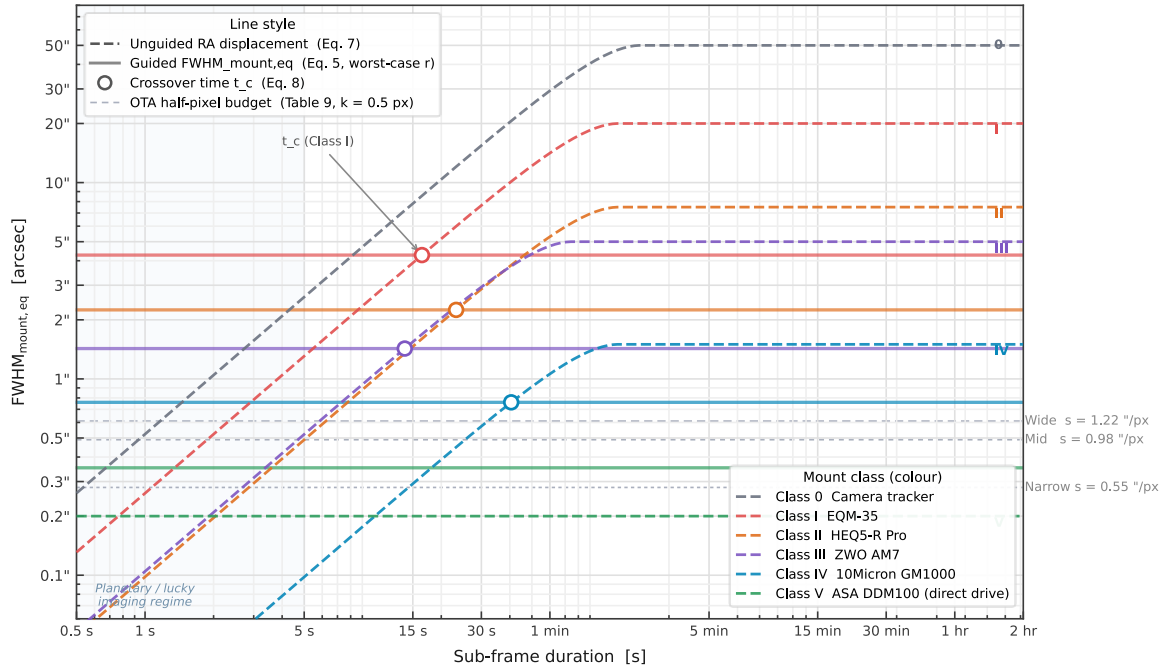
$\delta_{trail}(t)$ = Approximate RA displacement for a short exposure starting at the PE zero-crossing; $\delta_{trail}(t) = A_{PE} \times \sin(2\pi \times t / T_w)$ for $t \leq T_w/4$ (arcsec). If this displacement is large compared with the native star profile, it may approximate the RA broadening contribution. This should not be labelled as a general FWHM term.

T_w = Worm rotation period for worm-gear mounts; PE repetition period (s). Typical value: ~ 480 s for HEQ5-R Pro class. For harmonic-drive mounts, use the dominant period identified in the measured PE curve or guide log; a fixed worm period does not apply to harmonic-drive architecture.

Equation (7) identifies a crossover regime: at sufficiently short sub-frame durations, the PE has not accumulated enough displacement to exceed the irreducible guided noise floor, which is set by guide-star photon statistics, atmospheric tip-tilt faster than the guide update rate, and pulse quantisation. Equation (8) gives the crossover time t_c below which the approximate unguided displacement is smaller than the guided RA floor. The arcsin argument $(2.35 \times \sigma_{RA} / A_{PE})$ must be at or below 1.0 for a real solution to exist. If the argument exceeds 1.0, the simplified sinusoidal PE model does not predict a crossover within its valid range: the approximate $\delta_{trail}(t)$ remains below the assumed guided RA floor across the modelled interval. This does not by itself prove that unguided operation is superior, because unguided drift may also include non-sinusoidal PE components, polar misalignment, framing loss, and longer-time accumulation not captured by the simplified model Equation (7).

$$t_c = \frac{T_w}{2\pi} \cdot \arcsin\left(2.35 \frac{\sigma_{RA}}{A_{PE}}\right) \quad (8)$$

t_c = Crossover time; sub-frame duration below which the simplified unguided RA displacement estimate $\delta_{trail}(t)$ remains below the assumed guided RA displacement floor $2.35 \times \sigma_{RA}$. This is a phase-specific planning estimate, not a universal exposure-limit rule (see Equation 8 caveats).



$$\delta_{\text{trail}}(t) = A_{\text{PE}} \cdot \sin(2\pi t/T_w) \cdot \text{FWHM}_{\text{eq}} = (2.35/\sqrt{2}) \sqrt{\sigma_{\text{RA}}^2 + \sigma_{\text{DEC}}^2} \cdot t_c = (T_w/2\pi) \arcsin(\text{FWHM}_{\text{guided}}/A_{\text{PE}}) \quad \text{— Austronomic W7 journal, REV8 Eqs. 7, 5, 8}$$

Figure 2. FWHM_{mount} versus sub-frame exposure duration for six mount classes. Dashed curves: approximate unguided RA displacement from Equation (7), rising sinusoidally and plateauing at A_{PE} . Solid horizontal lines: guided FWHM_{mount,eq} from Equation (5) at worst-case DEC/RA ratio r per class (Table 7). Open circles: crossover times t_c from Equation (8). Below t_c , the unguided state produces less motion blur than the guided state. Three thin horizontal lines show the 0.5-pixel angular budgets for the wide, mid, and narrow telescope configurations in Table 9. Pure mount contribution; no seeing or diffraction components included. Dashed curves show angular displacement estimates (not FWHM); solid lines show guided FWHM floors per Equations (3) and (4).

For high-frame-rate planetary and lunar imaging, individual exposure times are typically short enough (0.01 to 5 seconds) that periodic error does not blur a single frame significantly, placing many configurations inside the crossover regime. Guiding is therefore not normally applied for individual-frame sharpness in this context; however, tracking and occasional corrections may still be useful to keep the target on the sensor and maintain framing during long capture sequences. Lucky imaging, which selects only the sharpest fraction of frames based on atmospheric quality, operates on this same regime and uses frame selection rather than

mount precision as the primary quality control mechanism.

For well-modelled encoder or direct-drive systems, the encoder noise floor is typically lower than the guided RA floor set by guide-star measurement noise, atmospheric tip-tilt, and pulse quantisation. In this scenario, unguided encoder-controlled operation may produce a lower mount PSF contribution than guided operation. This should still be validated against the specific payload, pointing model, site conditions, and exposure duration: flexure, model errors, refraction changes, wind, and pier movement can degrade unguided performance even on Class V systems.



Figure 2 plots mount-induced angular contribution versus sub-frame duration for all six mount classes, showing the approximate unguided RA displacement estimate (Equation 7, dashed rising curves) and the guided equivalent FWHM floor per axis (Equations 3-5, solid horizontal lines) with crossover points marked.

5.4 Polar Axis Alignment as a Performance Variable

The DEC/RA ratio r in Equations (5) and (6) is not a fixed mount property, it is a session-level variable controlled primarily by polar alignment quality. Perfect polar alignment means the mount's RA axis is exactly coincident with Earth's rotation axis. In that condition, Earth's rotation produces purely RA-direction apparent motion of any celestial target, with zero DEC component. The guide loop therefore issues no DEC corrections during the session, and DEC backlash never manifests. As a consequence, σ_{DEC} approaches σ_{RA} , r approaches 1.0, and RND approaches zero, regardless of the mount's backlash specification.

In practice, perfect polar alignment is unachievable. Electronic polar alignment routines (SharpCap polar alignment, ASIAIR, PoleMaster) achieve residuals of 2 to 10 arcminutes in the field. A 2-arcminute residual produces a DEC drift rate of approximately 0.5 to 1.0 arcseconds per minute depending on the target's hour angle and declination. Over a 5-minute sub-frame this accumulates to 2.5 to 5.0 arcseconds of DEC drift, requiring DEC corrections and engaging the DEC backlash mechanism. Polar alignment quality therefore sets a lower bound on achievable r : even with

zero mechanical backlash, a non-zero polar residual forces some DEC guiding activity.

The practical consequence for Class I and II mounts is that excellent polar alignment (residual below 2 to 5 arcminutes, achievable with electronic drift alignment) substantially reduces the frequency of DEC corrections and can lower r toward 1.0, reducing the elongation indicator toward zero without hardware change. However, polar alignment does not remove mechanical DEC backlash, structural flexure, wind response, cable drag, stiction, or guide-star seeing noise, these remain as residual error sources even with near-perfect alignment. Table 6 presents worst-case r estimates; the best-case (good alignment) scenario approaches $r = 1.0$ for all classes, subject to the remaining error sources listed.

- Polar alignment should be performed after the mount has thermally stabilised at the observing site (typically 20 to 30 minutes after setup), since thermal contraction of the tripod legs can shift the polar axis after initial alignment.
- Electronic drift alignment methods that use stellar centroid drift to compute the polar error vector are more precise than optical polar scope alignment in the field and achieve 2 to 5 arcminute residuals routinely.
- Re-aligning polar axis after each meridian flip is not necessary; the RA axis orientation is preserved through the flip. Only azimuth or altitude physical disturbance (such as moving the tripod) invalidates the polar alignment.
- For Class III harmonic-drive mounts ($r_{\text{worst}} = 1.2$), the improvement from good polar alignment is modest because the low backlash already keeps r near 1.0 without alignment improvement. The greatest



Table 7. Qualitative mechanical resilience classification by mount class. Settling time is the approximate time for vibration amplitude to decay below 0.1 arcseconds after a brief disturbance such as shutter actuation, mirror flip, or focuser adjustment. Values are indicative community-data ranges; actual performance depends on the specific telescope-mount-tripod assembly and site conditions. Mechanical resilience is a dynamic structural property governed by the natural frequency and damping ratio of the full system and cannot be reduced to a single numerical specification from the mount alone.

Class	Structural character	Approx. settling time	Wind resistance
0 Camera tracker	Light alloy head on ball head or simple alt-az; tripod-dependent	5-15 s (tripod-dependent)	Poor; small sail area but low stiffness
I Entry GEM	Light aluminium casting; standard aluminium tripod legs	4-8 s	Poor; susceptible to gusts above ~10 km/h
II Mid-range GEM	Medium-weight casting; stainless steel or aluminium tripod	3-5 s	Fair; adequate for calm conditions
III Upper-mid GEM / harmonic	Heavy casting or machined housing; rigid tripod or pier	2-4 s	Fair-Good; wind shielding recommended above 20 km/h
IV Premium GEM	Precision machined housing; massive tripod or dedicated pier	1-3 s	Good; handles moderate wind on solid pier
V Observatory / direct drive	Pier-mounted; massive machined assembly; active or passive damping	<1 s	Excellent; observatory enclosure recommended for maximum stability

benefit is realised for Class I and II worm-gear mounts with heavy DEC backlash.

5.5 Mechanical Resilience Classification

Mechanical resilience encompasses the mount's structural stiffness, vibration damping rate, and response to dynamic disturbances such as wind gusts, shutter actuation, and mirror slap. Unlike periodic error and guiding residuals, resilience cannot be reduced to a

reliable closed-form equation: it is governed by the natural frequency and damping ratio of the full telescope-mount-tripod assembly, which changes with every optical load configuration. The settling time (the time for vibration amplitude to decay below approximately 0.1 arcseconds after a disturbance) is the most directly relevant metric for astrophotography. Table 8 presents a qualitative classification by mount class; numerical values are indicative ranges from community reports and should be verified for any specific telescope-mount combination.

5.6 Mount Selection by Image Scale

Comparing this budget to the mount's guided RA residual σ_{RA} identifies the minimum class for that configuration: select the lowest class where σ_{RA} (from Table 6) is at or below $k \times s$, used here as an RMS planning tolerance consistent with PHD2's standard-deviation-like RMS reporting. The 0.5-pixel budget is a planning heuristic, not a hard boundary: 0.5 pixels or less is a strong target; 0.5 to 0.8 pixels is often usable in seeing-limited work where atmospheric blur dominates; above 1.0 pixel is likely visible in star shape unless exposure time, seeing, or scale masks it. Table 8 presents this selection matrix for the full range of image scales encountered in amateur astrophotography.

Table 8 presents image scale as the primary determinant of the angular tracking tolerance. The required mount class is then selected by combining that tolerance with payload capacity, moment arm, mechanical stiffness, guiding architecture, wind exposure,



Table 8. Mount selection guidance by image scale. The $k = 0.5$ pixel budget is a planning heuristic: 0.5 px is a strong target; 0.5-0.8 px often usable; above 1.0 px likely visible in star shape. Minimum class (unguided): lowest class whose A_{PE} amplitude is at or below the peak displacement budget ($k \times s$). Minimum class (guided): lowest class whose σ_{RA} from Table 6 is at or below $k \times s$ used as an RMS tolerance. These are not interchangeable criteria: the unguided budget is a peak displacement limit; the guided budget is an RMS limit; FWHM broadening requires a separate quadrature calculation if Gaussian assumptions hold. Image scale determines the tracking tolerance; payload, stiffness, wind exposure, and portability determine the final mount selection.

Image scale s (arcsec/px)	Typical regime	0.5 px angular budget (arcsec)	Min. class (unguided)	Min. class (guided)	Notes
> 3.0	Ultra wide-field; camera lens	> 1.50	Class 0	Class 0	Camera tracker sufficient for sub-minute exposures
1.5 - 3.0	Wide-field; short APO refractors	0.75 - 1.50	Class I	Class I-II	EQM-35 viable for wide-field with guiding
0.8 - 1.5	Mid-field; APO refractors 600- 1100 mm	0.40 - 0.75	Class II-III	Class II	HEQ5-R Pro or better; polar alignment critical
0.3 - 0.8	Narrow-field; long refractors, SCTs reduced	0.15 - 0.40	Not viable (PE too large)	Class III-IV	Harmonic or premium worm required; good polar alignment essential
< 0.3	Ultra-narrow; long SCT, planetary	< 0.15 (peak) / < 0.15 (RMS)	Not viable (PE >> budget)	Class IV-V	For planetary: use short unguided frames ($t < t_c$, Eq 8). Peak, RMS, and FWHM budgets are distinct, do not mix

portability constraints, and budget. Image scale alone is not sufficient to specify a mount: a configuration with a forgiving image scale but heavy payload, long moment arm, or exposed site may require a higher class than the tracking tolerance alone would indicate. Mount selection beginning with aperture or weight rather than image scale will, however, systematically under-invest at long focal lengths and over-invest at short ones.

6. Instrument Trade Study

The analytical component of the trade study quantifies the pixel consequence of a one-arcsecond angular tracking error for each standard telescope-camera configuration, and derives the maximum allowable angular error for half-pixel star displacement. By separating angular sensitivity from the mount-specific PE specification, the table enables any reader to estimate the PE pixel displacement for their own mount by multiplying the mount's PE amplitude by the sensitivity column.

Table 9 applies the image scale equation and Equation (2) to all ten standard telescope-camera configurations. The '1 arcsec error (px)' column gives the pixel displacement produced by any one-arcsecond angular error, regardless of its source (periodic error, polar misalignment, or guiding residual). The '0.5 px budget (arcsec)' column gives a practical angular-motion tolerance for mount-induced displacement from Equation (2) with $k = 0.5$. It is not a full delivered-image-quality budget: seeing, focus, optical aberration, and sampling also contribute to the measured star profile and are not included in this single-source budget. All image scale values are computed from $s = 206.265 \times p / f_{mm}$ using published instrument specs.

Table 9 demonstrates that the pixel consequence of any mount-induced angular tracking error grows in direct proportion to focal length. One arcsecond of angular error at the Askar 71F + ASI585MC Pro scale (row 1) displaces a star by 0.82 pixels; at the native EdgeHD 9.25 scale (row 10), the same angular



Table 9. Angular tracking-error sensitivity for all ten standard telescope-camera configurations. Image scale values verified from $s = 206.265 \times p / f_{\text{mm}}$ using published instrument specifications. The '1 arcsec error' column gives the pixel displacement from any one-arcsecond angular error from any source. The '0.5 px budget' column is the maximum mount-induced angular displacement that keeps the mount contribution below half a pixel. It is not a complete image-quality budget: seeing, focus, optical aberration, and sampling are not included in this displacement criterion.

Configuration (worked example)	f_{eff} (mm)	p (um)	s (arcsec/px)	1 arcsec error (px)	0.5 px budget (arcsec)
Askar 71F + ZWO ASI585MC Pro	490	2.90	1.22	0.82	0.61
Askar 91F APO + ZWO ASI585MC Pro	609	2.90	0.98	1.02	0.49
Askar 91F APO + ZWO ASI533MC Pro	609	3.76	1.27	0.79	0.64
Askar 103APO + ZWO ASI533MC Pro (native)	700	3.76	1.11	0.90	0.56
Askar 103APO + ZWO ASI533MC Pro (0.8x reduced)	560	3.76	1.38	0.72	0.69
Sky-Watcher Skymax 150 Pro + ZWO ASI585MC Pro	1800	2.90	0.33	3.01	0.17
Celestron EdgeHD 800 + ZWO ASI533MC Pro (0.7x reduced)	1422	3.76	0.55	1.83	0.28
Celestron EdgeHD 800 + ZWO ASI533MC Pro (native)	2032	3.76	0.38	2.62	0.19
Celestron EdgeHD 9.25 + ZWO ASI462MC (0.7x reduced)	1645	2.90	0.36	2.75	0.18
Celestron EdgeHD 9.25 + ZWO ASI462MC (native)	2350	2.90	0.25	3.93	0.13

error spans 3.93 pixels, a factor of nearly five. The 0.5-pixel budget column quantifies the maximum mount-induced displacement that keeps the mount contribution below half a pixel, it is not a full delivered-image-quality criterion, as seeing, focus, and optical aberration contribute separately. This displacement relationship is the direct consequence of Equation (2) for any mount-telescope-camera combination.

Combining Table 9 with the mount PE specifications in Table 5 yields the PE pixel displacement for any pairing. The EQM-35's typical amplitude of 15 to 30 arcseconds produces 12 to 25 pixels at the Askar 71F scale (row 1) and 45 to 90 pixels at the Skymax 150 Pro scale (row 6), while the budgets for those configurations are only 0.61 and 0.17 arcseconds respectively. Even the AM7's official ± 10 arcsecond peak PE specification, before guiding or encoder-based correction, far exceeds the half-pixel angular budget for every EdgeHD configuration. This does not mean the AM7 is unsuitable for these configurations; it means that suitability must be evaluated using measured guided RMS or measured corrected tracking, not raw PE alone. Autoguiding is a

system-level requirement for every configuration in Table 9 at the PE specifications in Table 5.

Periodic error can be measured on any equatorial mount using any imaging system capable of recording a bright star with sufficient signal-to-noise to track its centroid to sub-pixel accuracy over 10 to 20 minutes. The general method uses guiding software in logging-only mode: corrections are disabled so that star drift is recorded without intervention, producing an unmodified record of the mount's raw tracking error. The RA component of this record, plotted against time, reveals the PE signature illustrated in Figure 1, from which the worm period and the amplitude A_{PE} can be read directly for substitution into Equation (2). The objective of the experimental component is to confirm that the measured periodic-error pixel displacement agrees with the Equation (2) prediction within 10 percent, and that guided star quality in the imaging field is consistent with the guiding residual reported by the guiding software, after excluding frames affected by cloud, wind, or failed centroid tracking.



For the PE logging phase, a bright, isolated star near the meridian and above 45 degrees altitude serves as the guide target. A suitable PE measurement star from latitude -37.9 degrees South in southern Victoria is Sirius (alpha Canis Majoris; J2000: RA 06h 45m 09s, Dec -16 deg 42 min), visual magnitude -1.46, transiting at approximately 64 degrees altitude and accessible from November through March. Sirius provides extreme brightness for high-SNR centroid tracking; the single-star logging geometry is appropriate for raw PE measurement.

For the imaging validation phase, a star-rich field provides the FWHM and eccentricity measurements required to test whether guided tracking has reduced star trailing to the level predicted by Equation (2). A suitable target is the Sombrero Galaxy (M104, NGC 4594; J2000: RA 12h 39m 59s, Dec -11 deg 37 min), which transits at approximately 64 degrees altitude from latitude -37.9 degrees South, is accessible from March through May, and provides a background star field sufficient for plate solving, guided centroid tracking, and eccentricity measurement.

Suitable sites for this procedure are those with stable ground for the mount, an unobstructed southern horizon above 30 degrees, lower sky brightness than the surrounding suburbs, and protection from prevailing winds. General criteria for observers in the Melbourne region include distance from urban heat sources, elevation above surrounding terrain, and access at night. Specific site suitability should be assessed against current access conditions, sky-brightness maps, and session weather forecasts before each observing run.

6. Assemble the mount at the observing site and perform an initial electronic polar

alignment targeting a residual error below 10 arcminutes; this is sufficient that declination drift during the 15-minute PE logging run is small relative to the RA periodic error being measured.

7. Select Sirius or an equivalent bright star near the meridian and above 45 degrees altitude as the PE logging target; for the worked example configuration (Askar 71F + ZWO ASI585MC Pro), Sirius at transit satisfies both criteria from November through March.
8. Connect the guide camera to the mount's guide port and launch Open PHD Guiding 2 (PHD2, available at openphdguiding.org) or equivalent software capable of producing a timestamped RA residual log.
9. Enable centroid logging and disable all guide output, so the mount runs without correction and the recorded drift is the raw tracking error.
10. Acquire the guide star and confirm a signal-to-noise ratio above 20 at the selected frame rate; a threshold higher than the minimum for closed-loop guiding is warranted here because the mount is running without correction and any centroid noise adds directly to the recorded error signal.
11. Record for at least 15 minutes to capture at minimum one complete worm period; then export the RA error log, plot RA residual against time, and identify the dominant sinusoidal component to read off T_{worm} and A_{PE} for substitution into Equation (2).
12. Slew to M104 and plate solve a short test frame; calibrate the guide system and capture both an unguided sequence of 15-second sub-frames and a guided sequence of

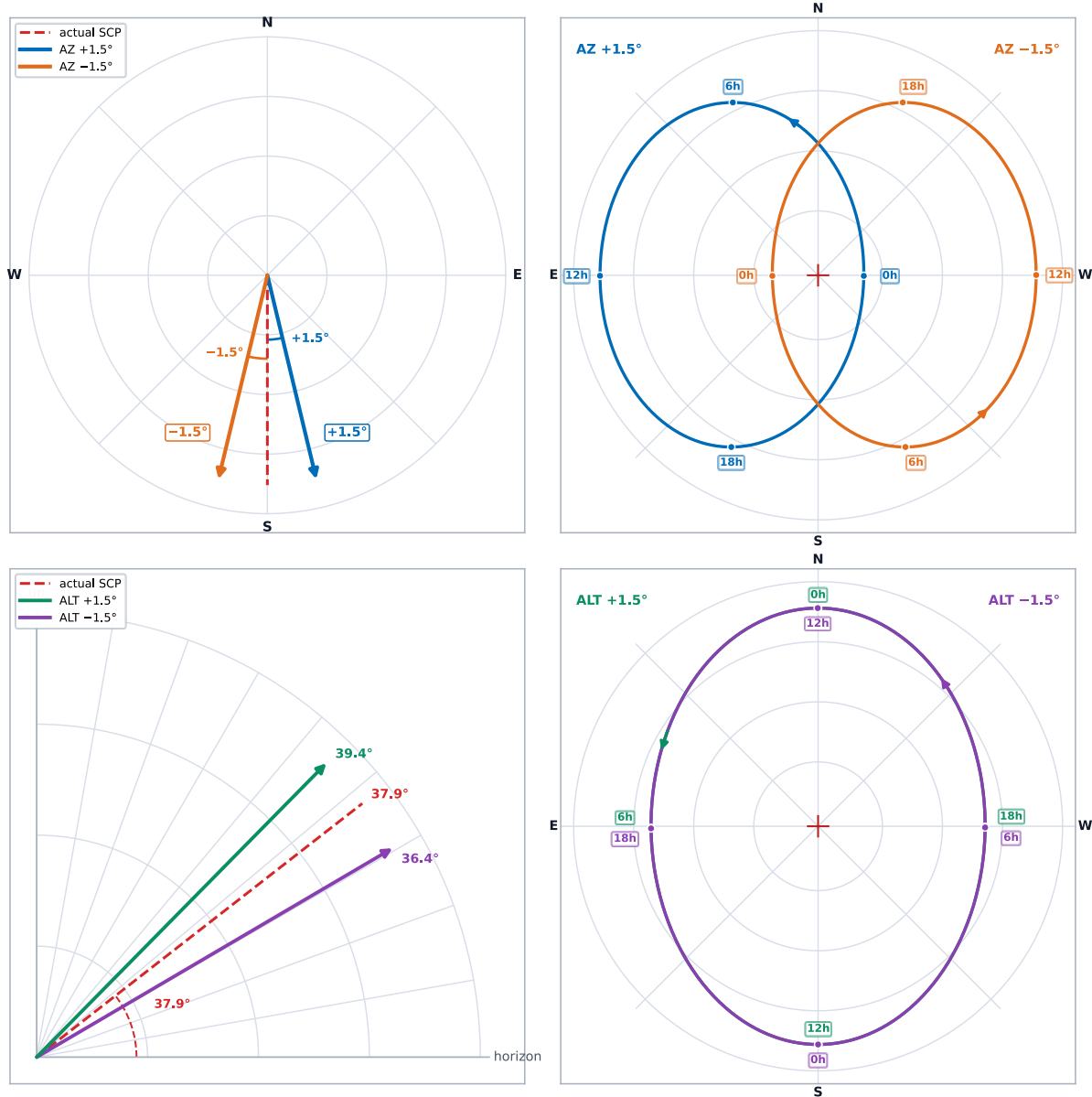


Figure 3. Polar-axis misalignment schematic for an equatorial mount in the southern hemisphere. The dashed red line marks the actual Southern Celestial Pole direction, while the coloured vectors show exaggerated $\pm 1.5^\circ$ azimuth and altitude alignment errors. The right-hand panels show the corresponding projected drift trajectories over one sidereal day, illustrating how azimuth and altitude errors produce distinct drift patterns in the field.

standard duration, measuring FWHM and star eccentricity from each set.

13. Compare the measured A_{PE} against Table 5 and the guided star FWHM improvement against the Equation (2) prediction; systematic exceedance of the

upper specification bound for PE indicates an off-balance condition, misadjusted worm mesh, or overloaded payload.

The primary validation criterion applies to the unguided PE log: the measured centroid



excursion amplitude from the RA drift log should agree with the Equation (2) pixel prediction (A_{PE} / s) to within 10 percent. This confirms that the mount's PE specification translates correctly to pixel-plane displacement at the system's image scale. For the guided imaging frames, the measured guide RMS (σ_{RA} , σ_{DEC} from PHD2) should be converted to imaging-camera pixels via σ / s and compared with changes in measured star eccentricity and FWHM from the M104 field. Delivered FWHM cannot be expected to match Equation (2) directly, as it also includes seeing, focus, and optical aberration. Persistent RA-aligned star elongation beyond 1.5 times the seeing-limited FWHM warrants review of balance, guide exposure, and algorithm aggressiveness [3].

All three mounts in the collection can be evaluated using this procedure sequentially at the same site and on the same targets. Simultaneous comparison of PE performance requires independent guide systems on each mount; sequential comparison under stable seeing conditions on one night provides a valid basis for ranking each configuration's tracking quality [5].

Figure 3 illustrates how polar-axis misalignment can be separated into azimuth and altitude components. The dashed red line marks the actual Southern Celestial Pole direction, while the coloured vectors show the $\pm 1.5^\circ$ alignment errors. The right-hand panels show how each error type produces a distinct projected drift trajectory over one sidereal day. Figure 3 separates the visual reference geometry from the resulting field drift, making the azimuth and altitude error signatures easier to compare.

7. Autoguiding and Residual Error Management

Even with a well-aligned mount and a drive system operating within its PE specification, residual tracking errors from polar misalignment, atmospheric refraction gradients, and mechanical flexure accumulate over multi-minute sub-frame exposures. Autoguiding addresses these residuals by making closed-loop corrections to the mount drive during the exposure. A guide camera, mounted on a separate guide scope or an off-axis guider, measures the centroid position of a selected guide star at a rate of several frames per second and issues correction pulses to the mount's guide port, keeping the star position within a programmable tolerance window expressed in guide-camera pixels [2].

Open PHD Guiding 2 (PHD2) is the standard autoguiding tool for equatorial mounts in amateur astrophotography. It measures residual drift in RA and declination, applies a correction algorithm with configurable aggressiveness and minimum-move thresholds, and outputs guide pulses through the mount's ST-4 or ASCOM interface. Calibration at the start of each session establishes the relationship between pulse duration and angular correction on sky, which varies with the guide scope focal length and the guide camera pixel scale. PHD2 reports RA and DEC RMS separately in arcseconds, together with the combined total RMS; in PHD2 convention, RMS is equivalent to standard deviation and approximately 68 percent of guide-star movements fall within the RMS value during steady-state guiding. The per-axis values σ_{RA} and σ_{DEC} are the inputs to Equations (3) and (4) from Section 5.2, and the total RMS is the



input to Equation (5) for the equivalent circular FWHM_{mount,eq}. The RND elongation indicator (Equation 6) can be used as a diagnostic: a persistent RND above 0.15 after achieving acceptable RA guiding suggests unresolved DEC backlash or polar alignment residual. This should be verified against measured eccentricity in the delivered image before use as an acceptance criterion (see Section 5.2 notes on the diagnostic heuristic status of RND) [5].

- The guide scope focal length should produce an image scale at the guide camera sufficient to detect star centroid shifts of 0.2 to 0.5 pixels per guide interval; a focal length between 150 and 400 mm with a small-pixel guide camera satisfies this criterion for the configurations in this series.
- The guide star must achieve a signal-to-noise ratio above 10 at the selected guide frame rate for reliable closed-loop correction; selection is typically automated by the guiding software from a stellar catalogue. Note that the SNR threshold for closed-loop guiding (10) is lower than the threshold for unguided PE logging (20 per the procedure in Section 6), because guiding averages the centroid over time whereas logging accumulates every measurement.
- All three mounts in the collection support the ST-4 guide port protocol and present an ASCOM interface for pulse guiding; either interface is compatible with PHD2 and equivalent guiding tools.

Autoguiding corrects errors that originate at or upstream of the guide star, but it cannot compensate for flexure between the guide optical path and the imaging optical path. Differential flexure, caused by gravitational sag in the mechanical connection between the

guide scope bracket and the imaging telescope, introduces a slow bias in the imaging field that accumulates during long exposures even when the guide star is held stationary. This effect is minimised by using an off-axis guider, which samples a star from the same optical path as the imaging camera, eliminating the independent mechanical degree of freedom between the two axes.

8. Forward Notes

The trade study quantifies the PE-limited pixel displacement for each configuration. The FWHM_{mount} framework in Section 3.4 extends this analysis to the guided regime, providing quantitative predictions of the mount's contribution to the delivered star PSF in both RA and DEC axes. A key result of the polar alignment analysis in Section 3.6 is that excellent polar alignment (residual below 2 arcminutes) reduces the DEC/RA ratio r toward 1.0 for any mount class, eliminating the star roundness (RND) penalty without any hardware change. For Class I and II mounts, this improvement alone can move the system from the visibly elongated regime (RND > 15 percent) to near-circular stars, making polar alignment quality the highest-leverage operational variable for these mount classes.

Session planning should account for thermal stabilisation (allow 20 to 30 minutes after setup before polar alignment), RA and declination balance verification, guide exposure duration selection that averages adequately over the predominant seeing frequency without introducing guide lag, and, for Class I and II mounts, electronic drift alignment targeting a residual below 5 arcminutes to keep r near the round-star threshold.



The interaction between mount performance and image scale formalised in Equation (2) has a direct consequence for how the series progresses. Weeks 5 and 6 established how the atmosphere limits image sharpness; Week 7 has established how the mount limits image stability. Both layers constrain the information reaching the sensor, but neither is a property of the sensor itself. Week 8 therefore opens the detector layer with a clean boundary: camera sampling and signal theory can be treated as operating on an image already shaped by atmosphere, optics, and mount. Mount selection should follow from image scale, not from telescope aperture alone, because the same angular PE error is more visible in pixels at longer focal lengths. The 0.5-pixel budget column in Table 9 gives the mount-induced displacement tolerance that any reader can apply to their own telescope-camera combination by substituting the relevant image scale.

Author Biography



Farzad Farajizadeh (Farzad Faradjizadeh) is an electrical engineer with a doctoral background and a multidisciplinary technical foundation spanning power systems, power electronics, electromagnetic compatibility, and related engineering domains. Through Austronomic, he explores astronomy from a systems-oriented perspective, with a particular emphasis on connecting physical principles, instrumentation, and engineering reasoning in a form accessible to motivated learners.

His broader objective with Austronomic is to develop a structured pathway into astronomy that moves beyond equipment-driven decision making and instead begins with conceptual clarity, technical rigor, and long-term system thinking. This work reflects his interest in building an educational framework in which astronomy can be approached not only as an observational pursuit, but also as a disciplined engineering and scientific practice.

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